6. Friday, April 11: **Basic set operations** (5.1)

Last Time: Open sentences $P(x)$ where $x \in U$ (variable, Universal set)

Set builder method $\{x \in U \mid P(x) \text{ is True}\}$.

Example: Consider the sets $A = \{-2, -1, 0, 1, \ldots, 5\}$ $B = \{2, 4, 8, 16, 32, \ldots\}$ (Roster with pattern)

In set builder notation, $A = \{n \in \mathbb{Z} \mid -2 \le n \le 5\}$, $B = \{n \in \mathbb{Z} \mid n = 2^k \text{ for some } k \in \mathbb{N}\}$

$= \{2^k \mid k \in \mathbb{N}\}$.

Set operations (short interlude, 5.1)

Definition 1. Let A and B be subsets of some universal set U.

1) The **union** of A and B is $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$ (\cup)

2) The **intersection** of A and B is $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$ (\cap)

3) The **difference** of A and B is $A \setminus B = \{x \in U \mid x \in A \text{ and } x \notin B\}$ (\setminus, set minus)

4) The **complement** of A is $A^c = \{x \in U \mid x \notin A\}$

Example: In the universal set $\mathbb{R}$, let $A=[-1,1]$ and $B=[0,2]$. Then

- $A \cup B = [-1,2]$
- $A \cap B = [0,1]$
- $A \setminus B = [-1,0)$
- $A^c = (-\infty,-1) \cup (1,\infty)$

Example: Let U be a universal set for a variable x. Take open sentences $P(x)$ and $Q(x)$, and denote their truth sets by A and B resp. Then:

1) $P(x) \vee Q(x)$ has truth set $A \cup B$.
2) $P(x) \wedge Q(x)$ has truth set $A \cap B$
3) $P(x) \wedge \neg Q(x)$ has truth set $A \setminus B$ (and $P(x) \wedge \neg Q(x) \equiv \neg(P(x) \rightarrow Q(x))$)
4) $\neg P(x)$ has truth set A^c

Quantifiers (2.4)

Recall: "$5x + 14 < 2$" is not a statement, but an open sentence (say the universal set is $\mathbb{R}$)

We can make it a statement if we quantify the variable x.

"There exists (quantifier) a real number (universal set) x (variable) such that $5x + 14 < 2$" (open sentence) is a statement

Definition: • The phrase "There exists" is called an existential quantifier

Notation: $\exists$ \exists

• An existential statement is one of the form

"There exists $x \in U$ such that $P(x)$" Notation: $(\exists x \in U)(P(x))$ (\exists)

where $P(x)$ is an open sentence with universe U.

• This statement is True if the truth set of $P(x)$ is non-empty, false otherwise

Example: "There exists a real number such that $x^2 = 2$."

Shortcut notation: $(\exists x \in \mathbb{R})(x^2 = 2)$

Note: This is True because $x^2 = 2$ has non-empty truth set

Definition: • The phrase "For every" is called a **universal quantifier**

Notation: $\forall$ \forall

• A **universal statement** is one of the form

"For every $x \in U$, we have $P(x)$" Notation: $(\forall x \in U)(P(x))$

• This statement is **True** if the truth set for $P(x)$ is all of U; false otherwise.

Example: "For every real number x, we have $x^2 \geq 0$"

Shortcut: $(\forall x \in \mathbb{R})(x \geq 0)$

Note this is True, because the truth set of "$x^2 \geq 0$" is all of $\mathbb{R}$.

Example: "For every real number x, we have $x^2 \geq 1$" $(\forall x \in \mathbb{R})(x^2 \geq 1)$

False, truth set is not all of $\mathbb{R}$, it does not contain 0.

Statements with Multiple quantifiers

Example 3: Write the next statement in symbols:

"For every natural number x, there exists a natural number y such that $x<y$"

$$(\forall x \in \mathbb{N})(\exists y \in \mathbb{N})\ (x<y)$$

This is true: given $x \in \mathbb{N}$, we have $x+1 \in \mathbb{N}$ which satisfies $x < x+1$.

- Write the following in English:

$$(\exists y \in \mathbb{N})(\forall x \in \mathbb{N})\ (x<y)$$

"There exists a natural y such that for every natural x we have $x<y$".

This is false; there is no natural number larger than all naturals.

- Are these statements the same? No!

So order of quantifiers matters.

Negating Quantifiers

Example 4: Negate the statement "Everybody loves math".

$(\forall x \in U)$ (x loves math) Where U = set of all people.

Incorrect answer: "Everybody does not love math"

$(\forall x \in U)$ (x does not love math)

Correct answer: "There exists some person who does not love math"

$(\exists x \in U)$ (x does not love math)

- Negation of Universal statement: $\neg(\forall x \in U)(P(x)) \equiv (\exists x \in U)(\neg P(x))$

Example 5: Negation of "At least one person loves hiking" $(\exists x \in U)$ (x loves hiking)

is "No one loves hiking" $(\forall x \in U)$ (x does not love hiking)

- Negation of Existential statement: $\neg(\exists x \in U)(P(x)) \equiv (\forall x \in U)(\neg P(x))$

Example 6: Negate "For every real number x, there exists a real number y such that $x > y$"

Answer: In symbols: $(\forall x \in \mathbb{R})(\exists y \in \mathbb{R})(x > y)$

To negate:

$$\neg(\forall x \in \mathbb{R})(\exists y \in \mathbb{R})(x > y) \equiv (\exists x \in \mathbb{R})(\neg(\exists y \in \mathbb{R})(x > y))$$

$$\equiv (\exists x \in \mathbb{R})(\forall y \in \mathbb{R})(\neg(x > y))$$

$$\equiv (\exists x \in \mathbb{R})(\forall y \in \mathbb{R})(x \leq y)$$

"There exists some $x \in \mathbb{R}$ such that for every $y \in \mathbb{R}$ we have $x \leq y$".